Comparison of Sub 6 GHz and mmWave Wireless Channel Measurements at High Speeds

F. Pasic, D. Schützenhöfer, E. Jirousek, R. Langwieser, H. Groll, S. Pratschner, S. Caban, S. Schwarz and M. Rupp

Presenter: Faruk Pasic

Technische Universität Wien

Institute of Telecommunications

Introduction

- increasing demand for high data rate transmission

[1] J. Larson, "Korean patterns of mobile internet use," Dec. 2013.

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- mmWave bands allow for significantly higher data rates

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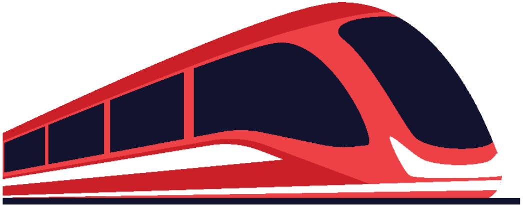
- high speed train scenarios well investigated in realistic environments



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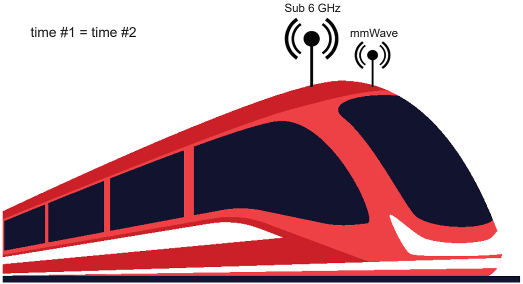

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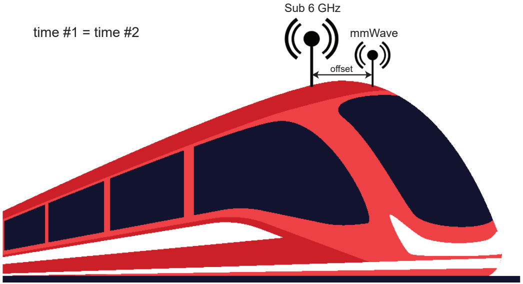

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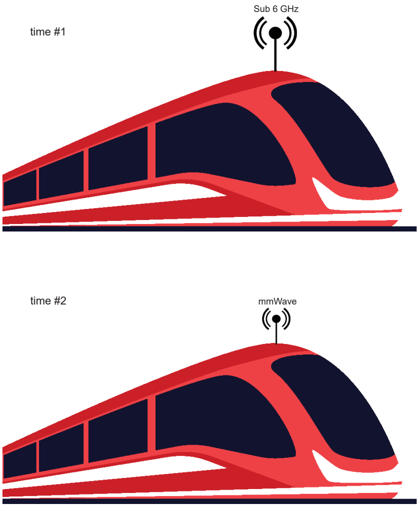

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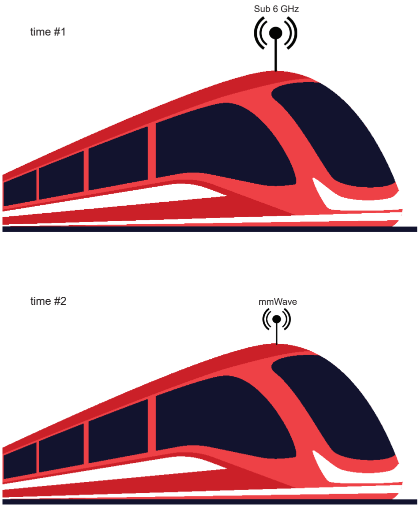
  - same time $\rightarrow$ position offset
  - same position $\rightarrow$ repeatability lost
  - different small-scale fading behavior



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[2] F. Pasic, S. Pratschner, R. Langwieser, D. Schützenhöfer, E. Jirousek, H. Groll, S. Caban, and M. Rupp
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  - testbed hardware and the methodology proposed in [2]

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- **results of the measured wireless channel compared in terms of delay-Doppler function**

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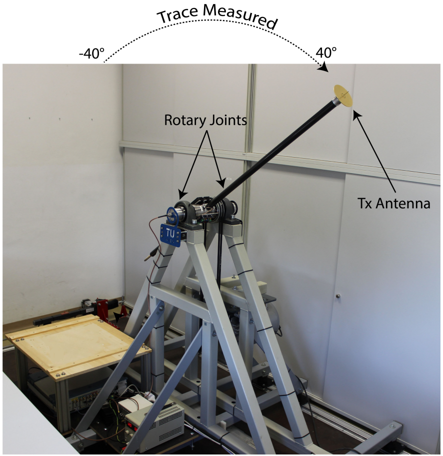
- rotary unit
  - trigger unit
  - rotary joints
  - repeatable measurements up to 400 km/h in controlled environment



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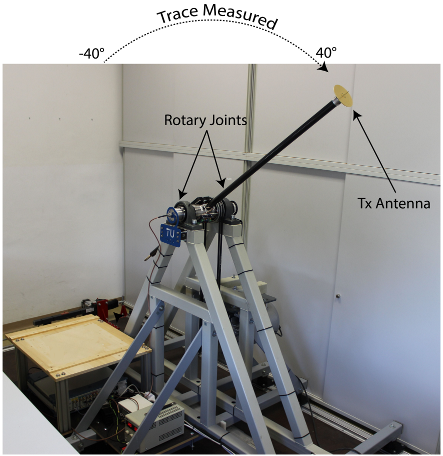
  - rotating Tx and statical Rx
  - perform channel measurements along the same trace for different frequency bands



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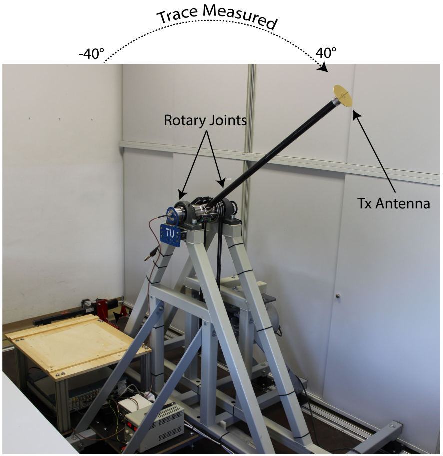
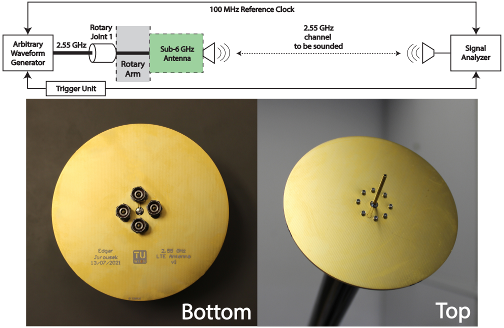
  - rotating Tx and statical Rx
  - perform channel measurements along the same trace for different frequency bands
- problem: rotary joint limitation
  - maximum signal frequency of 12.4 GHz
  - setup modification required



Testbed Architecture - Sub 6 GHz case

- direct transmission is possible



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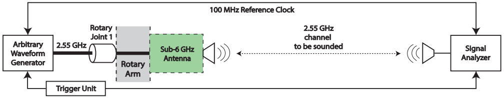
- Tx signal generated at a center frequency $f_{\mathrm{RF}} = 2.55\,\mathrm{GHz}$



Testbed Architecture - Sub 6 GHz case

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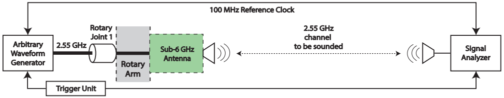
- transmitted by vertical dipole antenna



Testbed Architecture - mmWave case

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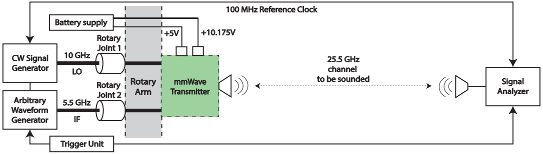

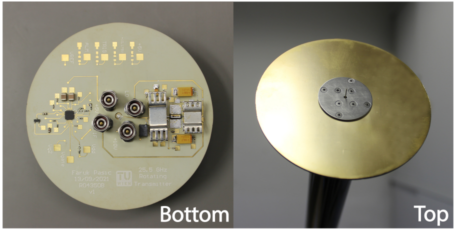

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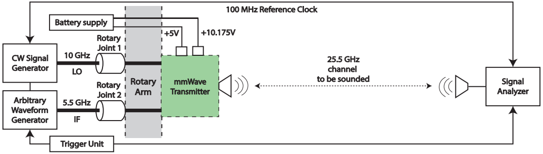
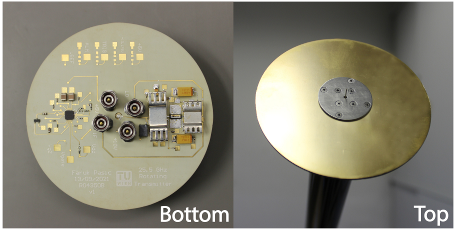
- frequency up-conversion by means of lightweight mmWave Tx to $f_{\mathrm{RF}} = 25.5\,\mathrm{GHz}$





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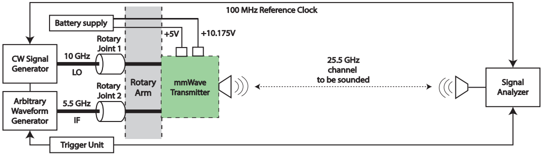

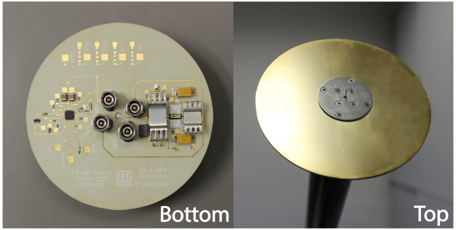

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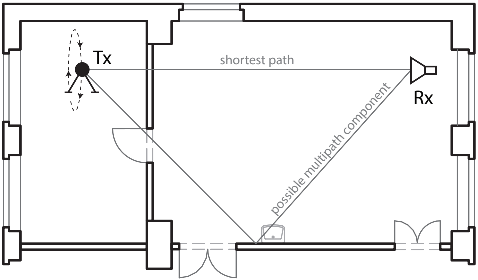
  - moving Tx antenna and static Rx antenna
  - static fading environment



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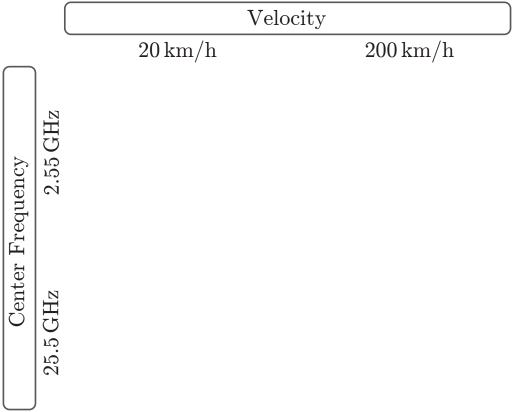
- measurement procedure
  - channel sounding using OFDM
  - four combinations of low or high speed and sub 6 GHz or mmWave center frequency



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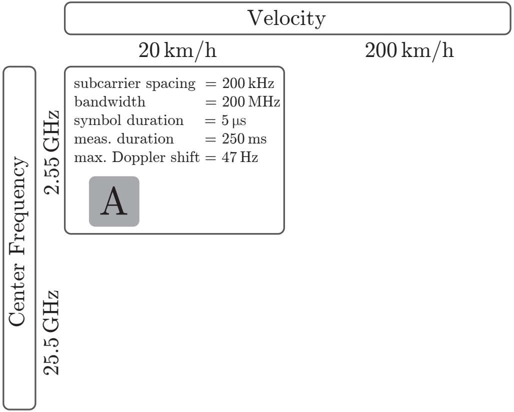

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Center Frequency \ Velocity	20 km/h	200 km/h
2.55 GHz	subcarrier spacing = 200 kHz bandwidth = 200 MHz symbol duration = 5 µs meas. duration = 250 ms max. Doppler shift = 47 Hz A	subcarrier spacing = 2 MHz bandwidth = 200 MHz symbol duration = 500 ns meas. duration = 25 ms max. Doppler shift = 472 Hz B
25.5 GHz		

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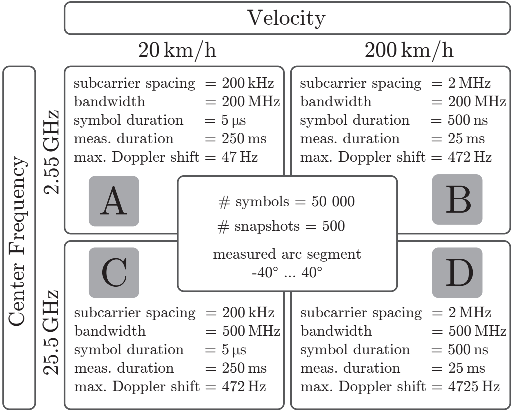

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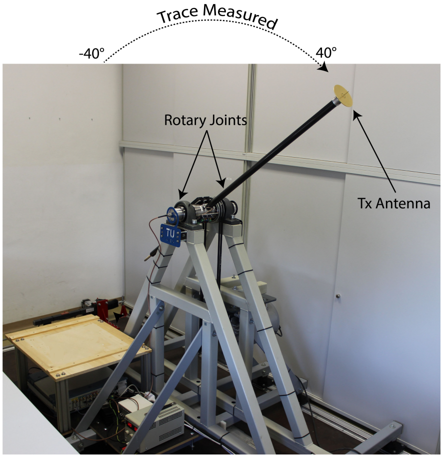

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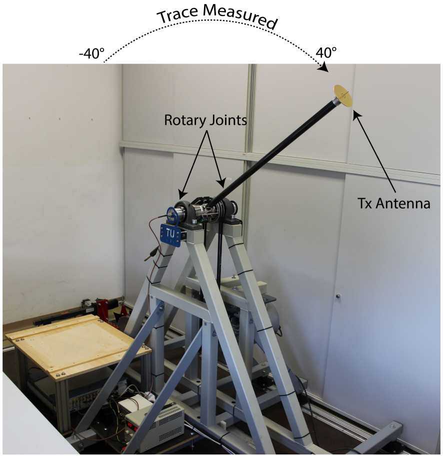
- result: time-variant channel transfer function



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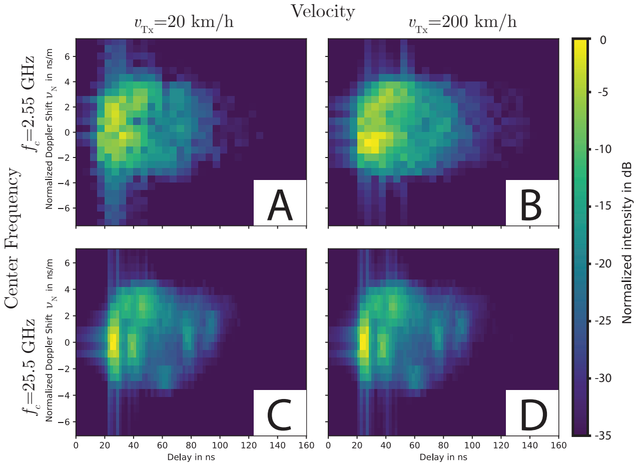
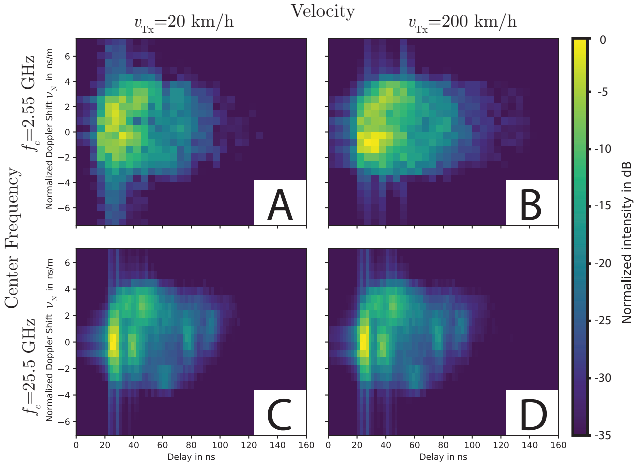
Measurement Results

- results in terms of delay-Doppler functions



Measurement Results

- results in terms of delay-Doppler functions
- normalizations performed:
  - normalized Doppler Shift $\nu_\mathrm{N} = \frac{\nu}{f_\mathrm{c}\, v_\mathrm{Tx}}$
  - intensity normalized to 0 dB



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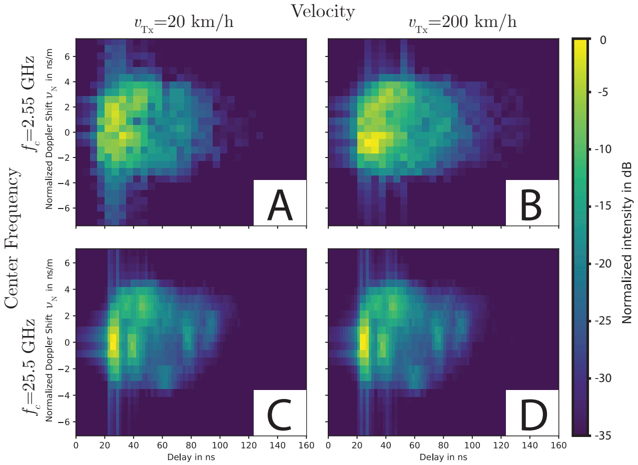
- normalizations performed:
  - normalized Doppler Shift $\nu_{\mathrm{N}} = \frac{\nu}{f_{\mathrm{c}} v_{\mathrm{Tx}}}$
  - intensity normalized to 0 dB
- key observations for delay-Doppler functions:
  - invariance with regard to the transmitter's velocity
  - different for two different center frequencies



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- normalized delay-Doppler function invariant with regard to the transmitter's velocity
  - **measurements to compare sub 6 GHz and mmWave in high-mobility scenarios are possible**

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- normalized delay-Doppler function invariant with regard to the transmitter's velocity
  - **measurements to compare sub 6 GHz and mmWave in high-mobility scenarios are possible**
- measured delay-Doppler functions look different despite same trace and velocity
  - **scattering environment is frequency selective**

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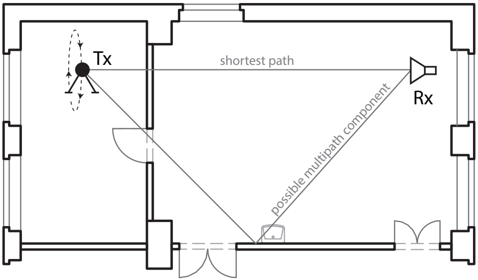
Why Rotating Tx and Statical Rx ?

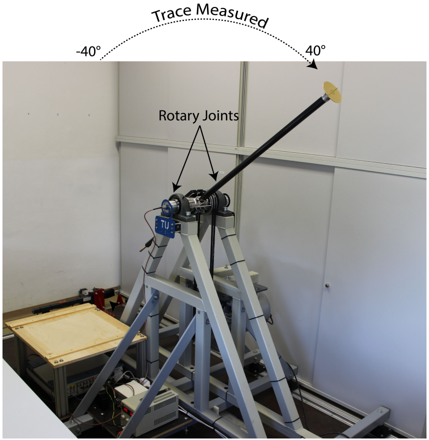
- sub 6 GHz case
  - no limitation
  - rotary unit can be used either as Tx or Rx
- mmWave case
  - frequency conversion at the end of the arm
  - Rohde & Schwarz Signal Analyzer R&S FSW67 with satisfying noise figure available



Replacement of Rotary Joints

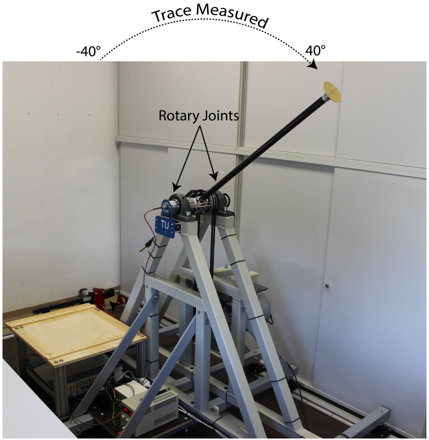
- advantages
  - direct signal transmission at mmWave band

- drawbacks
  - high insertion losses
  - limited rotational speeds
  - replacement of cables and RF connectors
    → high costs



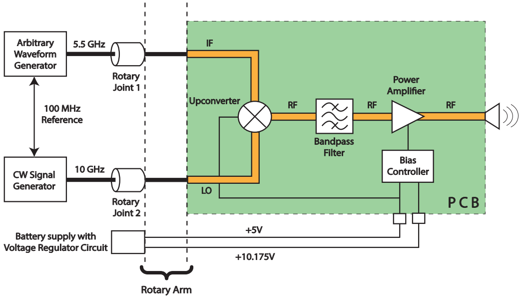
Rotary Unit

- repeatable, closed-loop, off-line-processed measurements at very high velocities
  - 1500 revolutions per minute (up to 400 km/h)
  - aluminum alloy arm (length of 1m)
  - semi-rigid coaxial radio-frequency cables
- repeatability
  - keep the surroundings constant
  - no objects move
  - repeat a position with accuracy of 100 μm
- controllability
  - change average SNR (Tx power)
  - change velocity (rotation speed)
- measurement repetition rate
  - realistic train scenario $\rightarrow$ few measurements until the train has passed the transmitter
  - many measurements in short time-spans



mmWave Transmitter

- high acceleration forces
  - compact size and light weight
  - no bulky connectorized RF modules
- up-converter
  - MAMF-011024
  - $f_{RF} = 2f_{LO} + f_{IF}$
  - output contains additional spectral products
- bandpass filter
  - B259MC1S
  - 25 – 26 GHz bandpass frequency range
  - suppress unwanted spectral products
- power amplifier
  - HMC994AM5E
  - output power level too low
  - uses active bias controller (HMC980LP4E)



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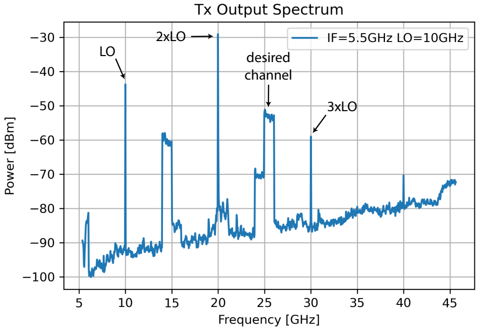

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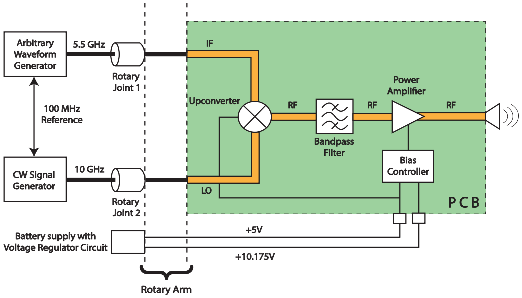
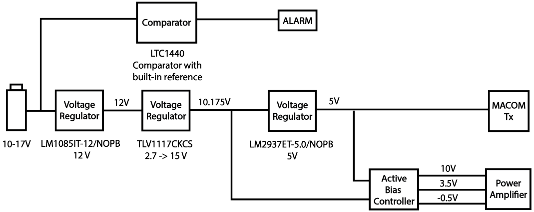


Battery Supply

- $+5$V and $+10.175$V DC power supply required by transceiver and active bias controller
- no rotary joints left to transmit the DC supply during the rotation
- possible solution: DC supply provided by battery power supply



Trigger Unit

- used for time synchronization
- uses rotational encoder
- trigger measurement at precisely defined position of rotary arm
- 960 possible angle positions (accuracy of 300 μm)
- signal of rotary encoder decoded by counter and comparator to form trigger signal
- trigger signal fed to Tx-Generator and Rx-Analyzer

Frequency Synchronization

- Tx and Rx interconnected with 100 MHz reference
- not feasible in drive-by measurements

Local Scattering Function

- non-stationary time-variant frequency transfer function obtained as result

- assumption: channel is locally stationary
  - 6λ of motion (250 snapshots for sub 6 GHz and 25 snapshots for mmWave case)
  - entire frequency range

- calculate local scattering function according to [3] and [4]:
  - group non-stationary measured frequency transfer function into more local stationary frequency transfer functions
  - for each of them perform FFT to obtain local stationary delay-Doppler functions
  - average delay-Doppler functions over number of taken local stationary groups

[3] M. Hofer et al., "Wireless vehicular multiband measurements in centimeterwave and millimeterwave bands" in 32nd Annual International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC), Helsinki, Finland, Sep. 2021.
[4] L. Bernado et al., "Delay and Doppler spreads of nonstationary vehicular channels for safety-relevant scenarios", IEEE Transactions on Vehicular Technology, vol. 63, no. 1, 2014.